

Sahar Iqbal Malik

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Professional Summary

Software Engineering undergraduate (FAST-NUCES) with hands-on experience building full-stack web applications using the MERN stack and modern React/Next.js tooling. Experienced in API development, authentication, database integration, and shipping production-ready UI. Additional exposure to applied machine learning (NLP, optimization, scikit-learn) through academic and project work. Strong product mindset with experience collaborating cross-functionally and delivering measurable user-impact outcomes.

Skills

Programming Languages: JavaScript (ES6+), TypeScript, Python, SQL, C++, Java

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive UI, SPA Architecture, shadcn/ui

Backend: Node.js, Express.js, REST APIs, API Integration, JWT Authentication, Sessions

Databases: MongoDB, MySQL, Database Design, CRUD Operations

AI/ML: scikit-learn, NLP, Genetic Algorithms, Optimization, Model Evaluation, RAG (Retrieval-Augmented Generation), Vector Databases

Tools & Deployment: Git/GitHub, Postman, VS Code, Netlify, Vercel

Concepts: Agile/Scrum, SDLC, Authentication/Authorization, Debugging, Clean Code

Education

FAST-NUCES, Islamabad

BS in Software Engineering | 2022 – 2026

Relevant Coursework: Data Structures, Web Development, Databases, Software Engineering, Machine Learning

Westminster Academy, Islamabad

A Levels – Maths, Physics, Computer Science (A B B) | 2020 – 2022

SAT: 1360

O Levels | 2018 – 2020

Experience

Product Management Intern – Nayatel

Islamabad | June 2025 – July 2025

- Collaborated with cross-functional teams (engineering/design) to define user-focused features aligned with product strategy.
- Supported roadmap planning and sprint execution using Agile/Scrum workflows.
- Conducted competitive analysis and gathered user feedback to inform prioritization and improvements.

Product Analyst – CodingKey

Islamabad | Jan 2024 – Aug 2024

- Improved product feature performance across backend and frontend, contributing to ~15% retention uplift.
- Performed data analysis using SQL and Python to guide product logic and roadmap decisions.
- Coordinated between engineering and design teams to deliver functional releases under deadlines.

Founder & Owner – Captain Hook (E-commerce Side Hustle)

Remote | *June 2023 – Present*

- Built and scaled a direct-to-consumer business to 150+ orders in the first year.
- Managed end-to-end operations including inventory, customer support, and storefront workflows.
- Applied product thinking to improve customer journey through Instagram + WhatsApp commerce.

Management Intern – *Specialist Care Clinic*

Islamabad | *June 2020 – Aug 2020*

- Managed high-volume front desk operations and improved patient intake efficiency.
- Reduced average wait time by ~20% through workflow and scheduling optimization.

Projects

Aegis (FYP) – *React, TypeScript, Tailwind*

An enterprise security platform combining ML-based intrusion detection, pentest advisory modules, and a RAG agent for threat intelligence. Integrated real-time IDS alerts, automated security recommendations, and an AI-powered Q&A system that retrieves context from security documents.

DigiXTech – *Next.js 15, TypeScript, Tailwind, shadcn/ui*

Developed a production-ready Next.js starter template designed for SaaS dashboards and internal admin tools. Implemented reusable UI components, responsive layouts, and scalable structure for rapid development.

AI Agent for Disaster Volunteer Allocation – *Python, Streamlit*

Created an optimization-based allocation engine and dashboard to distribute volunteers/resources across zones. Designed interactive UI and logic for real-time configuration and scenario testing.

Secure End-to-End Encrypted Messaging Platform – *React, Node.js, Web Crypto*

Built E2EE messaging and file-sharing system using AES-256-GCM and ECDH (P-256) key exchange. Implemented ECDSA signatures, nonce management, and threat-model-aligned security flows.

CineMatch – *Python, NLP, Genetic Algorithms*

Built a recommendation system combining sentiment analysis and evolutionary optimization for personalized results. Evaluated performance and improved recommendation relevance using experiment-based tuning.

Custom CNN – *TensorFlow / Keras, scikit-learn, Python*

Developed a chest X-ray pneumonia detection CNN (TensorFlow/Keras) with end-to-end ML pipeline - achieving 85.4% accuracy and 0.916 AUC - including ablation experiments with augmentation, dropout, and L2 regularization.

Leadership & Activities

Co-Vice Head, Girls Volleyball – FAST-NUCES (*Aug 2024 – September 2025*)

Led a 10+ member team, fostering team coordination and discipline.

Vice President, Management – Sustainable Development Society

Organized large-scale awareness events and led operational logistics.

Marketing Team Member – NASCON 2024

Contributed to one of Pakistan's premier student-led tech and sports fests.